

Mohamed Eldagla

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Education

German International University in Cairo (GIU)

October 2025 – Present

Master's in Data Science, Minor in Software Engineering

New Administrative Capital, Egypt

Advised by Dr. Caroline Sabty

- *Featured Coursework:* Analytics on the Cloud, Computer Graphics and Visualizations, Data Engineering and Visualizations, Software Engineering Studio

Egypt-Japan University of Science and Technology (E-JUST)

2021 – 2025

BSc. in Computer Science (AI and Data Science)

Alexandria, Egypt

- *Featured Coursework:* Advanced NLP (CMU), Natural Language Processing, CS 188: Introduction to AI (UC Berkeley), Computer Vision, Neural Networks, New Trends in AI, Mathematics of Data Science, Probability & Statistics

Skills

Languages: Python, C/C++, JavaScript | **Toolkits:** TensorFlow, PyTorch, OpenCV

Data & Tools: HuggingFace Transformers, PEFT/LoRA, vLLM, Spark, Kafka, SQL, MongoDB, Pandas, Git, Docker, L^AT_EX

Professional Experience

Egyptian Cabinet Information and Decision Support Center

July 2025 – September 2025

AI Engineer Intern

New Administrative Capital, Egypt

- Led development of a Prime Minister-sponsored RAG-based rumor detection system for the “Ask Egypt” platform.
- Built scalable pipelines for automated verification of news content from trusted governmental and media sources.

Research Experience

Fatima Institute for Global AI Research

April 2026 – Present

Pre-doctoral Fellow — Curiosity-Driven Learning in LLM Agents & Multilingual AI Safety

Remote

- Studying learning-progress rewards for LLM agents across inference, SFT, and GRPO under Alfonso Amayuelas (Stanford HAI).
- Built a noisy-TV test environment to compare surprisal, learning progress, and information gain under irreducible uncertainty.
- Auditing multilingual, multi-turn specification-following with Tasmiah Tahsin Mayeesha (University of North Texas).
- Selected as 1 of 67 fellows from 700+ applicants.

PRISM AI Safety Research Fellowship

June 2026 – Present

Research Fellow — Reward Model Interpretability

Remote

- Interpreting personalized reward-model bases with Rachel Freedman (UC Berkeley CHAI) and collaborators across Columbia, CMU, Stanford, AIMS, and Intel.
- Selected as 1 of 50 fellows from 800+ applicants.

MIT AI Safety Fellowship

June 2026 – August 2026

Research Fellow

Remote

- Selected from 300+ applicants for a summer fellowship on alignment, scalable oversight, and mechanistic interpretability.

Université Paris-Est Créteil (UPEC)

June 2025 – August 2025

Data Scientist Intern (NLP)

Remote

- Selected as 1 of 30 from 1,300+ applicants for a master's-level AI research internship as an undergraduate.
- Built an LLM-based event discovery system (TinyLlama 1.1B) over 50K+ records from 100+ sources; applied ensemble knowledge distillation using Jais 13B, Gemma 3, and TunBERT.

Publications

Reward Basis Decomposition for Interpreting Preference Heterogeneity. *Under review, NeurIPS Workshop.* Rachel Freedman (UC Berkeley). Separates shared and individual reward directions.

Uncertainty, Not Concealment: A Mechanistic Reappraisal of Sycophancy in Vision-Language Models. *Under review, NeurIPS Workshop.* Probing and steering reveal uncertainty on cross-modal conflicts.

Specification Drift in Multilingual, Multi-Turn LLM Interactions. *In preparation.* Audits specification-following across multilingual, multi-turn conversations.

The Curious Agent: Can Intrinsic Curiosity Guide LLM Exploration?. *In preparation.* Alfonso Amayuelas (UCSB). Tests curiosity-driven exploration across inference, SFT, and GRPO.

Mokhiamar, M. O., Mahmoud, A., Eldagla, M. I., et al. (2025). **Blockchain-Integrated Multi-Modal LSTM-CNN Fusion for High-Precision Epileptic Seizure Detection from EEG Signals.** *Knowledge-Based Systems*, 323, 113703. [Paper]

Elshamy, A. A., Metwaly, H. A., Eldagla, M. I., et al. (2025). **EEG Spectrogram Augmentation Using Transformer and Quantum-Inspired Generative Models for Epilepsy Classification.** *JAC-ECC 2025.* [Paper]

Teaching Experience & Service

German International University in Cairo (GIU) — Faculty of Informatics

Winter 2025 – Spring 2026

Teaching Assistant

New Administrative Capital, Egypt

- Taught Data Engineering and Visualizations (50 students), Big Data & NoSQL (80), Programming I (120), and Databases (154); led lectures/labs, projects, office hours, and grading.
- Covered data pipelines and visualization, MongoDB, CassandraDB, SparkML, Kafka, Hadoop, Python, SQL, relational algebra, and database design.

Code in Place (CS106A)

April 2026 – July 2026

Section Leader (Mentor) — Stanford University

Remote

- Led weekly sessions for Stanford's free introductory Python course, serving 16,000+ learners worldwide.

Digital Egypt Cubs Initiative (DECI)

July 2023 – December 2024

Instructor — Udacity

Egypt

- Taught Python, data structures, and algorithms to 40 students aged 12–17 in weekly four-hour, activity-based lessons.

Google Developer Student Clubs (DSC)

October 2021 – May 2023

Core Team Member & CS50x Instructor — E-JUST

Alexandria, Egypt

- Taught C++, data structures, and algorithms to 50+ undergraduates; delivered CS50x lectures to 100+ students and led coding sessions.

United Nations — COP27

October 2022 – November 2022

Information Communication Technology Specialist

Sharm El-Sheikh

- Collaborated with MCIT and UN engineering teams to support the COP27 mobile application used during the international climate conference.

Honors & Awards

Full Fellowship for MSc in Data Science

2025

German International University in Cairo (GIU)

Cairo, Egypt

- Full-tuition MSc Data Science fellowship for academic excellence and research potential.

Academic Excellence Recognition

2025

Egypt-Japan University of Science and Technology

Alexandria, Egypt

- Final-semester 4.00 GPA (ranked 1/120); featured in university newsletter for academic excellence.
- Graduated in the top 10%; received a 70% merit scholarship.

SAS Curiosity Cup: Data Science Global Competition

2024

Top 15 Finalist out of 107 Global Participants

Egypt